

JAGIELLONIAN UNIVERSITY MEDICAL COLLEGE
DOCTORAL SCHOOL OF MEDICAL AND HEALTH SCIENCES

WHAT IS THE OPTIMAL GRAFT CHOICE FOR PRIMARY ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION SURGERY?



A systematic review and network meta-analysis

A dissertation submitted in partial fulfillment of the requirements for the award of the degree of:
Doctor of Philosophy (Ph.D.)

submitted by:

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[Network meta-analysis of autografts and allografts](#)

Pairwise meta-analyses:

- Bone-patellar tendon-bone autograft vs. Hamstring tendon autograft

Bone-patellar tendon-bone autograft vs. Quadriceps tendon autograft

Hamstring tendon autograft vs. Quadriceps tendon autograft

What is the optimal graft choice for primary anterior cruciate ligament reconstruction surgery?

A systematic review and network meta-analysis of randomised controlled trials.

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Abstract

Objective: The objective of the review was to synthesise the results between the most common graft types for primary anterior cruciate ligament reconstruction surgery. In this review, four autologous grafts (bone-patellar tendon-bone, hamstring, quadriceps, and peroneus longus tendon) and two allogenic grafts (Achilles tendon and tibialis anterior tendon) were included and compared in terms of patient-reported outcome measures, antero-posterior translation and posterolateral knee stability, and graft failure.

Design: Systematic review of randomised controlled trials

Methods: Data sources: The literature search was initially performed on July 29, 2026 for all six graft types from database sources: PubMed/MEDLINE, Embase, Web of Science Core Collection, and Cochrane Library. **Study eligibility criteria:** Sixty-nine randomised controlled trials that compared the results of at least one of the specified outcome measures between at least two of the specified graft types were included in the qualitative review. Forty-five studies reported sufficient data on outcome measures to be included in the quantitative review. **Synthesis methods:** Two-arm and multi-arm study results underwent network meta-analysis of standardized mean differences (SMDs) and odds ratios (ORs) for continuous and dichotomous variables. Direct and indirect comparisons between the four autologus grafts and the two allogenic grafts were summarised with a regression analysis of covariates as predictors: follow-up duration (months) and short, medium, and long-term follow-up duration, mean age and male to female gender ratios.

Results:

Conclusion:

Protocol registration: PROSPERO ([CRD42024592549](#)) and Cochrane Reviews (COCHRANEEMD-2026-00379)

INTRODUCTION

Rationale

The anterior cruciate ligament (ACL) is responsible for approximately 70% of antero-posterior and postero-lateral stability of the knee joint [citation]. A complete rupture of the ACL requires surgical reconstruction of the ligament in order to restore the native biomechanical properties of the joint. Although extra-articular procedures gained initial popularity for performing ACLR, they have since been phased out as an auxiliary procedure, called lateral extra-articular tenodesis (LET) and thus, intra-articular procedures currently dominate the ACLR scene. Currently, the most common grafts utilized for such intra-articular ACLR include autologous tendons and allogenic tendons, and less commonly, synthetic sources (1). Despite the comprehensive scope by which ACLR has been studied, with numerous systematic reviews and meta-analysis papers on the topic, a consensus has not been made regarding the optimal graft choice (1).

Autologous grafts—harvested from the tendons of the patients’ own muscles, include the patellar tendon, the hamstring tendon, the quadriceps tendon, and the peroneus longus tendon. The patellar tendon—otherwise known as the bone-patellar tendon-bone (BPTB) graft—is currently the gold-standard and most common choice for ACL reconstruction surgery. The hamstring tendon (HT), includes single, double, triple and quadruple-bundled semitendinosus with or without gracilis tendon, and has been associated with outcomes that closely resemble those of the BPTB. The quadriceps tendon (QT), whether harvested with or without the patellar bony attachment has been shown to be a non-inferior choice for ACLR (2). The peroneus longus tendon (PLT) can be harvested with or without peroneus brevis tenodesis.

Allogenic grafts—harvested from exogenous sources—include the allogenic BPTB, Achilles tendon (AT), and the tibialis anterior tendon (TA). Currently, the only utilized synthetic or artificial sources of ACL is the Leeds-Keido graft.

Objectives

Although there are many graft harvesting choices from which the surgeon must preferentially decide from, there are no comprehensive studies that compare the outcomes of all of these aforementioned grafts. Many randomized controlled trials (RCTs) exist that compare the most common choices, such as BPTB, HT, and QT; however, only a few studies compare autografts with allografts. Thus, a network meta-analysis was conducted in order to produce a comprehensive set of outcome data that summarizes the comparative results between all of the grafts inclusive of all contemporary surgical graft harvesting options.

Commented [dongwkim1]: BPTB
HT (choose most common abbreviations)

- SB
- DB
- 4ST (semitendinosus only)
- 4STG (semitendinosus with gracilis)
- 6STG

QT

- B-QT
- S-QT

PLT

- PLT
- PLT+PBT

Commented [dongwkim2]: Revise to make it sound a bit better.

METHODS

Protocol and registration

The current review adhered strictly to the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and the extension for Network Meta-Analyses (PRISMA-NMA) (3). The review protocol was developed and registered on the International Prospective Registry of Systematic Reviews (PROSPERO) under the registration number: [CRD42024592549](#). The review protocol was submitted and accepted on Cochrane Database of Interventional Reviews under registration number: COCHRANEEMD-2026-00379.

Eligibility criteria

The inclusion and exclusion criteria for selecting studies are shown in Supplementary Table 1.

Population: Studies conducted on skeletally-mature, adult patients after complete rupture (grade III sprain) of the anterior cruciate ligament were included. Skeletal maturity was determined to be a study in which the mean age of all participants was greater than or equal to 18 years old. Studies which were conducted on pediatric or adolescent patients, or in which the mean age of all participants was less than 18 years were excluded.

Intervention: Studies in which the patients underwent primary anterior cruciate ligament reconstruction were included. Studies in which the patients underwent revision or repair of the anterior cruciate ligament were excluded.

Comparator(s): Studies in which the primary anterior cruciate ligament was reconstructed using the bone-patellar tendon-bone autograft, hamstring tendon autograft, quadriceps tendon autograft, peroneus longus tendon autograft, Achilles tendon allograft or tibialis anterior tendon allograft were included in the study. Autografts must have been harvested from the ipsilateral limb of the patients, and the allografts were included regardless of their prior γ -irradiation status prior to transplantation. Studies that performed the reconstruction using any other graft were excluded from the study.

Outcome(s): Studies that reported on the IKDC subjective knee form (IKDC-SKF) score, Lysholm knee scoring scale, Tegner activity scale score, instrumented laxity measured as the side-to-side difference in anteroposterior tibial translation by KT-1000/2000 or Rolimeter, pivot-shift test, Lachman test, and graft failure (including graft re-rupture, clinical failure, or revision surgery) were included in the study. Studies that did not report on at least one of the specified outcomes were excluded from the quantitative review.

Study design: Studies that were randomised controlled trials (RCTs) were included. All other study designs, including but not limited to non-randomised controlled trials, prospective and retrospective cohort studies (comparative and non-comparative), case-control studies, and case series and/or reports were excluded.

There were no restrictions placed on length of follow-up or year of publication. No restrictions were placed on language for retrieval of full-text reports; however, only studies written in English were considered for inclusion. Only studies published in peer-reviewed journals were included in the review.

The studies were grouped by the graft harvested for reconstructing the anterior cruciate ligament. There was a total of 12 different graft types, which were grouped into six subgroups for the purposes of the network meta-analysis. The groups were formed according to the muscle of which the tendon was harvested from.

Table 1 Graft types included in study

No.	Subgroup	Intervention	Full name and description
1	BPTB	BPTB	Bone-patellar tendon-bone autograft.
2		2-ST	Double-stranded semitendinosus hamstring autograft.
3	HT	4-ST	Quadruple-stranded semitendinosus hamstring autograft.
4		4-STG	Quadruple-looped semitendinosus and gracilis autograft.
5		6-STG	Six-stranded semitendinosus and gracilis autograft.
6	QT	B-QT	Quadriceps tendon with bone autograft.
7		S-QT	All soft-tissue quadriceps tendon autograft.
8	PLT	PLT	Peroneus longus tendon autograft.
9		PLT + PBT	Peroneus longus tendon autograft with peroneus longus brevis tenodesis.
10		AT	Achilles tendon allograft.
11	TA	TA	Tibialis anterior tendon allograft.
12		TP	Tibialis posterior tendon allograft.

Commented [dongwkim3]: PRISMA Item #24 (a-c)

Register the pre-print on OSF and the other one.

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☑ Specify inclusion and exclusion criteria for the review.

☑ How the studies were grouped for syntheses.

Information sources

The literature search was conducted using the following three databases: PubMed/MEDLINE, Embase, and Web of Science Core Collection. The Cochrane Central Register of Controlled Trials (CENTRAL) and reference lists of systematic reviews and network meta-analysis review articles on the topic were used as a source for citation search. The database and register searches were performed up to and including July 29, 2026. Reference/citation search was performed on August 21, 2026.

Search strategy

The inclusion criteria for intervention, comparators and study design were used to construct the search strategy for each database. The inclusion criteria for intervention included the following search terms, "anterior cruciate ligament", "anterior cruciate ligament reconstruction", "ACL", and "ACLR", which were combined by the Boolean operator, "OR". For comparators, "bone-patellar tendon-bone", "patellar tendon", "BPTB", "hamstring tendon", "semitendinosus", "gracilis", "quadriceps tendon", "QT", "peroneus longus", "PLT", "Achilles tendon", and "tibialis anterior" were used. Study design was used to filter out "randomised controlled trials" and "clinical trials" using publication type field tag and filters. The full search strategy for each of the three database/registry searches is shown in Supplementary Table 2. The full search strategy for PubMed/MEDLINE (shown below) was translated into Embase, Web of Science Core Collection and Cochrane Central Register of Controlled Trials (CENTRAL) search syntax.

PubMed/MEDLINE

1. "anterior cruciate ligament"[MeSH Term]
2. "anterior cruciate ligament reconstruction"[Title/Abstract]
3. "anterior cruciate ligament reconstruction"[Title/Abstract]
4. "ACL"[Title/Abstract]
5. "ACLR"[Title/Abstract]
6. #1 OR #2 OR #3 OR #4 OR #5
7. "bone-patellar tendon-bone"[Title/Abstract]
8. "patellar tendon"[Title/Abstract]
9. "BPTB"[Title/Abstract]
10. #7 OR #8 OR #9
11. "hamstring"[Title/Abstract]
12. "hamstring tendon"[Title/Abstract]
13. "semitendinosus"[Title/Abstract]
14. "gracilis"[Title/Abstract]
15. #11 OR #12 OR #13 OR #14
16. "quadriceps"[Title/Abstract]
17. "quadriceps tendon"[Title/Abstract]
18. "QT"[Title/Abstract]
19. #16 OR #17 OR #18
20. "peroneus"[Title/Abstract]
21. "peroneus longus"[Title/Abstract]
22. "fibularis longus"[Title/Abstract]
23. #20 OR #21 OR #22
24. "achilles"[Title/Abstract]
25. "achilles tendon"[Title/Abstract]
26. #24 OR #25
27. "tibialis"[Title/Abstract]
28. "tibialis anterior"[Title/Abstract]
29. "tibialis posterior"[Title/Abstract]
30. #27 OR #28 OR #29
31. #10 OR #15 OR #19 OR #23 OR #26 OR #30
32. "randomized controlled trials"[Publication Type]
33. "randomized controlled trials"[Title/Abstract]
34. "randomised controlled trials"[Title/Abstract]
35. #32 OR #33 OR #34
36. "review"[Publication Type]
37. "systematic review"[Publication Type]
38. "meta-analysis"[Publication Type]
39. "review"[Title/Abstract]
40. "systematic review"[Title/Abstract]
41. "meta-analysis"[Title/Abstract]
42. #36 OR #37 OR #38 OR #39 OR #40 OR #41

Commented [dongwkim5]: PRISMA Item #6

Specify all databases, registers, etc. searched.

Specify the date of last search.

43. #35 NOT #42
44. #6 AND #31 AND #43

There were no limitations on date of publication, language, or any other filters other than publication type in the database and registry searches.

Study selection process

One reviewer (D.W.K.) performed the literature search on July 29th 2026. Database search resulted in 622, 317, 531 and 234 records for PubMed/MEDLINE, Embase, Web of Science Core Collection, and Cochrane Central Register of Controlled Trials (CENTRAL), respectively, for a total of 1,704 records. Nine hundred and ninety-seven duplicate records were removed. The titles and abstracts of 707 records were screened, and 103 full-text reports were sought for retrieval. All 103 reports were retrieved and underwent full-text screening. Thirty-four reports were excluded from the review during full-text screening for the following reasons for exclusion: two reports were not randomised controlled trials, 17 studies did not report on at least one of the specified outcomes, two did not compare at least two of the specified graft types, nine were duplicates of the same study, and four reports were written in a language other than English. Sixty-nine studies were included in the qualitative review and 45 studies were included in the quantitative review. The study selection process is depicted in Figure 2.

Data collection process

Data extraction was performed using spreadsheets created on Microsoft Excel®. Data extraction was performed by one independent reviewer. Although data collection forms were created and piloted, the extraction was performed directly on Microsoft Excel® spreadsheets for consistency in methods between the reviewers. Data extraction for study results included the subgroups, data type (continuous and dichotomous), sample size, effect size (mean or event), median, range (minimum and maximum), interquartile range (first and third quartiles), and standard deviations. Furthermore, for pivot-shift and Lachman outcomes, the sample size and event (0, 1+, 2+, and 3+) were extracted. For graft failure, the sample size and event (re-rupture, clinical failures, and revision) were extracted.

Data items

Data collection involved extracting (1) study characteristics (total participants, including total number of male and female participants), the sample size at last follow-up, (including sample size of male and female participants), number lost due to follow-up, study design, and the level of evidence), (2) recruitment details (country, department, institution, period of enrollment (start and end dates of the recruitment period), (3) covariates (mean age and standard deviation, follow-up duration in months (minimum, mean and standard deviation) and the male to female ratio taken from the sample size for each gender), and (4) outcomes (IKDC subjective knee form scores, Lysholm knee scoring scale, Tegner activity scale, instrumented laxity, pivot-shift, lachman, graft failure (including graft re-rupture, clinical failure, and revision ACLR).

Study risk of bias assessment

The risk of bias assessment was performed using the Cochrane Revised Risk of Bias Assessment Tool (R.o.B.2) for randomised controlled trials. Each randomised controlled trial was assessed for risk of bias in five domains: bias in randomisation, bias from deviation from treatment intention, bias in measurement of the outcome, reporting or selection bias, and bias in missing outcomes. An overall risk of bias was given according to the R.o.B 2 tool with a designation for each study derived from the algorithm for determining whether a study has ‘High risk’, ‘Low risk’ or ‘Some concerns’ regarding bias.

Effect measures

The effect measures extracted for continuous outcomes were the mean and standard deviation. If a study did not report the mean, but reported the median and range (minimum and maximum) or interquartile range, the mean was estimated using transformation methods for estimating the sample mean from the sample size, median, and range or from the sample size, median, range, and interquartile ranges. For continuous outcomes, the standardised mean difference (SMD) and standard error of the SMD were calculated between the two treatment groups (i.e., graft types) of two-arm and multi-arm studies. For dichotomous outcomes, the odds ratio (OR) and the standard error of the OR was calculated.

Network meta-analysis (synthesis methods)

Commented [D6]: Two independent reviewers needed.

Commented [dongwkim7]: Get 2nd reviewer. Add co-authors.

Commented [dongwkim8]: Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.

Commented [dongwkim9]: Get 2nd reviewer. Complete the piloted forms to make it easier for the 2nd reviewer. Try using something other than the jupyter ipywidgets forms (TypeForm, etc.) that is easy to collaborate with.

Commented [dongwkim10]: 10a. List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.

10b. List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.

Commented [dongwkim11]: Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.

Commented [dongwkim12]: Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.

The effect sizes of individual studies were grouped by graft type and synthesised into summary estimates of standardised mean difference (SMD) and the 95% confidence interval., or odds ratio (OR) and the 95% confidence interval for continuous and dichotomous outcomes respectively, using network meta-analysis of random effects. The outcome measures were categorised into three separate dimensions: (1) subjective patient satisfaction, (2) objective knee stability and (3) graft failure. Subjective patient satisfaction following primary ACLR involved patient-reported outcome measures (PROMs), such as IKDC subjective, Lysholm and Tegner scores. Objective measures of knee stability included instrumental laxity, pivot-shift and Lachman tests. Graft failure included graft re-rupture, clinical failure, and revision surgery.

RESULTS

Study selection

Identification of studies included search of databases PubMed/MEDLINE, Embase, Web of Science Core Collection, and the Cochrane Central Register of Controlled Trials (CENTRAL) registry, which resulted in 622, 317, 531 and 234 records respectively, for a total of 1,704 records. Nine hundred and ninety-seven duplicate records were removed, leaving titles and abstracts of 707 records to be screened. Six hundred and four records were excluded during title and abstract screening. One hundred and three full-text reports were sought and retrieved and screened. Thirty-four reports were excluded from the review for the following reasons: two reports were not randomised controlled trials, 17 studies did not report on at least one of the specified outcomes, two did not compare at least two of the specified graft types, nine were duplicates of the same study, and four reports were written in a language other than English. Fifty-nine studies were included in the qualitative review and 45 studies were included in the quantitative review. The study selection process is depicted in Figure 2.

Study characteristics

The study characteristics recorded were the total number of participants (including total number of male and female), sample size at last follow-up, number of participants lost during follow-up, study design, and level of evidence. Covariates for regression analysis included follow-up duration in months (minimum, mean and standard deviation), age (mean and standard deviation), and gender (male to female ratio).

Risk of bias in studies

Risk of bias assessment was performed using Cochrane’s Risk of Bias Assessment Tool (Ro.B. 2).

Reporting biases

Certainty of evidence

Geometry of the network

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Results of individual studies

The results of individual studies are shown in Supplementary Tables 3-9.

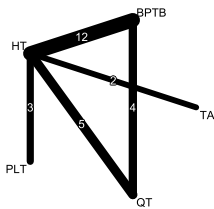
Synthesis of results

Patient-reported outcome measures

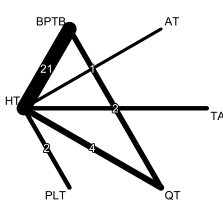
Twenty-six studies reported on IKDC Subjective Knee Form scores (4–29), 35 studies reported on Lysholm Knee Scoring Scale scores (4,6,8–14,16,17,19,21,22,25–28,30–44), and 27 studies reported on Tegner activity scale scores (4,6,7,14–16,19,25,28,30–33,37–42,44–47).

Fig 1. Network graphs of patient-reported outcome measures (A-C)

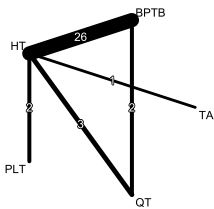
A. IKDC Subjective



B. Lysholm



C. Tegner

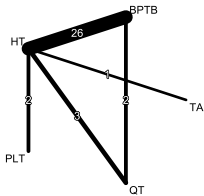


Objective measures of knee stability

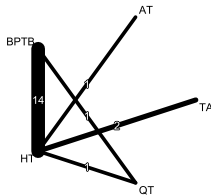
Forty-eight studies reported on instrumented laxity (1,4,5,11,14–20,24,25,28–30,32–34,36–39,41,43,47–55), 30 studies reported on pivot-shift test (1,5,6,10,11,14–16,18,19,33,35,40,41,43,44,46–48,50,51,53,54,56–58), and 26 studies reported on Lachman test (6,10,14,17,19,32,33,35,37,39–41,44,47,48,51,52,54,56,58).

Fig 2. Objective measures of knee stability

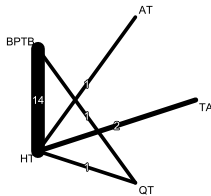
A. Instrumented laxity



B. Lachman test[†] ($\geq 1+$)



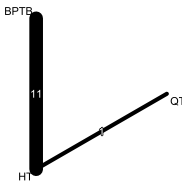
C. Pivot-shift test ($\geq 1+$)



Graft failure

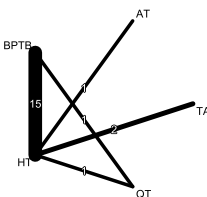
Graft failure was defined as either (1) graft re-rupture confirmed by MRI or arthroscopy, (2) clinical failure defined as $\geq$ grade 2+ or 3+ on Lachman and/or pivot-shift tests, or (3) revision ACLR surgery. The sum of the three nominal data was recorded as the number of events in a dichotomous dataset.

A. Graft failure

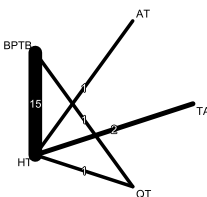


Clinical failure

B. Pivot-shift test[‡]



C. Lachman test[§]



[†] Lachman test $\geq 1+$ (3-5 mm) grade was considered an event.

[‡] Pivot-shift test $\geq 2+$ (clunk) was considered a clinical graft failure.

[§] Lachman test $\geq 2+$ (5-10 mm) grade was considered an event.

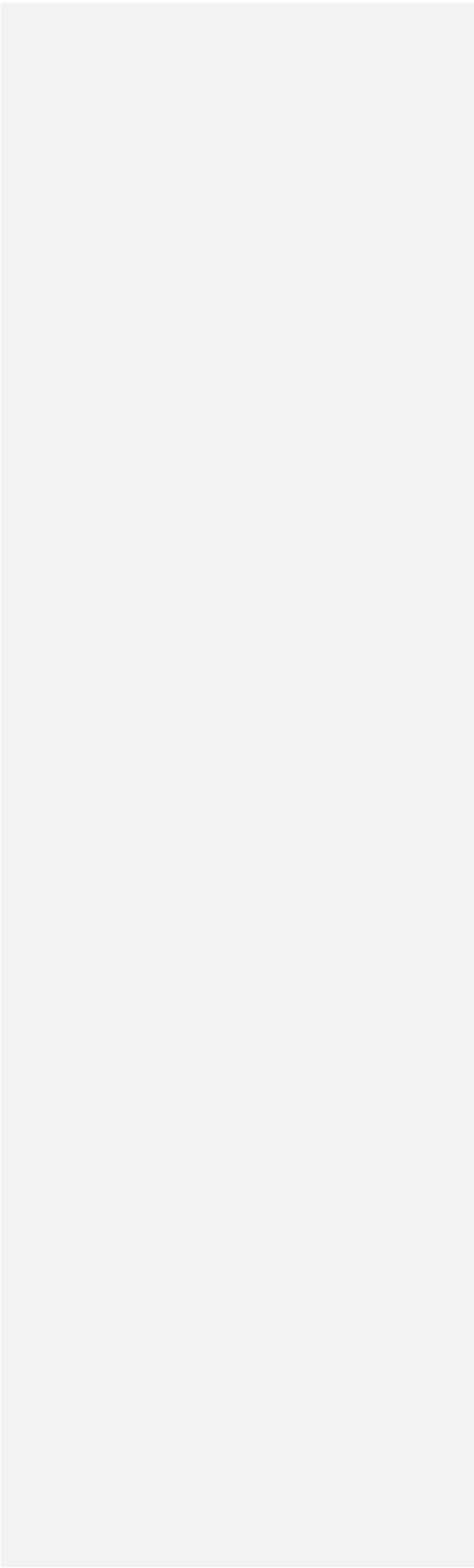
DISCUSSION

Summary of evidence

Limitations

Conclusions

Funding



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Data availability statement

The datasets used for analyses in this review were made publicly accessible on Open Science Framework (DOI: [10.17605/osf.io/ssnda](https://doi.org/10.17605/osf.io/ssnda)) and Harvard Dataverse ([10.7910/DVN/NGIW6E](https://doi.org/10.7910/DVN/NGIW6E)).

TABLES AND FIGURES

Tables

Table 1 Eligibility criteria for inclusion and exclusion of studies

Table 2 Database search strategies

Table 3 Graft types included in study

No.	Subgroup	Intervention	Full name and description
1	BPTB	BPTB	Bone-patellar tendon-bone autograft.
2		2-ST	Double-stranded semitendinosus hamstring autograft.
3		4-ST	Quadruple-stranded semitendinosus hamstring autograft.
4		4-STG	Quadruple-looped semitendinosus and gracilis autograft.
5	HT	6-STG	Six-stranded semitendinosus and gracilis autograft.
6		B-QT	Quadriceps tendon with bone autograft.
7		S-QT	All soft-tissue quadriceps tendon autograft.
8		PLT	Peroneus longus tendon autograft.
9	PLT	PLT + PBT	Peroneus longus tendon autograft with peroneus longus brevis tenodesis.
10		AT	Achilles tendon allograft.
11	TA	TA	Tibialis anterior tendon allograft.
12		TP	Tibialis posterior tendon allograft.

Table 10 Graft failure

Table 4 Characteristics of included studies

Table 4 IKDC Subjective Knee Form

Table 5 Lysholm Knee Scoring Scale

Table 6 Tegner Activity Scale

Table 7 Instrumented Laxity

Table 8 Pivot-Shift Test

Table 9 Lachman Test

Table 4 Characteristics of included studies.

ID	Study, Author et al., year	Arms		Covariates							Outcomes						
		Intervention 1	Intervention 2	Age, Mean ± SD	Sample size, n			Follow-up duration			Continuous				Dichotomous		
					Total, n	Male, n	Female, n	Months	Years	Category	IKDC subjective	Lysholm	Tegner	Instrumented Laxity	Pivot-shift	Lachman	Graft failure
1	Ageberg et al., 2009	BPTB	HT	30.0 ± 5.1	36	27	9	36	3	Medium	X	X	✓	X	X	X	X
2	Aglietti et al., 2004	BPTB	HT	25.0 ± 5.1	120	92	28	24	2	Medium	X	X	X	✓	✓	✓	X
3	Ahldén et al., 2009	BPTB	HT	28.0 ± 5.1	47	32	15	88.5	7.4	Long	X	✓	✓	✓	X	✓	X
5	Anderson et al., 2001	BPTB	HT	20.6 ± 5.9	103	68	35	35.9	3	Medium	✓	X	X	✓	✓	X	X
6	Anz et al., 2023	BPTB	HT	28.5 ± 6.4	20	7	13	12	1	Short	X	X	X	X	X	X	✓
7	Aune et al., 2001	BPTB	HT	26.0 ± 5.8	72	40	32	24	2	Medium	X	X	X	✓	X	X	X
9	Barenius et al., 2010	BPTB	HT	34.0 ± 7.9	153	89	64	96	8	Long	✓	✓	X	✓	✓	✓	X
10	Barié et al., 2020	BPTB	QT	30.6 ± 8.1	60	34	26	123	10.3	Long	✓	✓	✓	✓	X	X	✓
11	Beynnon et al., 2002	BPTB	HT	29.2 ± 7.0	56	31	25	36	3	Medium	✓	X	✓	✓	✓	✓	X
12	Bi et al., 2018	HT	PLT	28.5 ± 7.6	124	65	59	30	2.5	Medium	✓	X	X	✓	✓	X	✓
13	Björnsson et al., 2016	BPTB	HT	27.5 ± 6.3	147	95	52	197.3	16.5	Long	✓	✓	✓	✓	X	X	X
14	Bottoni et al., 2015	HT	TA	29.1 ± 6.6	97	84	13	120	10	Long	✓	X	✓	X	X	X	X
18	Butt et al., 2024	HT	PLT	28.5 ± 8.3	60	59	1	60	5	Long	✓	✓	X	X	✓	✓	X
19	Calvert et al., 2025	HT	QT	28.8 ± 5.6	112	56	56	24	2	Medium	✓	X	X	✓	X	X	X
23	Ejerhed et al., 2003	BPTB	HT	29.0 ± 4.6	66	46	20	24	2	Medium	X	✓	✓	✓	X	✓	X
25	Eriksson et al., 2001	BPTB	HT	27.0 ± 7.9	116	0	0	6.7	0.6	Short	X	X	X	✓	✓	X	X
26	Feller & Webster, 2003	BPTB	HT	26.1 ± 9.7	65	47	18	36	3	Medium	✓	X	X	✓	✓	✓	X
30	Gifstad et al., 2013	BPTB	HT	27.0 ± 6.0	77	0	0	84	7	Long	X	✓	✓	✓	X	✓	X
32	Guglielmetti et al., 2021	BPTB	HT	24.9 ± 6.0	62	42	20	24	2	Medium	✓	✓	X	X	✓	✓	X
33	Harilainen et al., 2006	BPTB	HT	.0 ± 6.7	79	0	0	60.6	5.1	Long	✓	✓	✓	✓	✓	X	X
34	Heijne & Werner, 2010	BPTB	HT	29.5 ± 5.4	68	36	32	24.7	2.1	Medium	X	X	✓	✓	✓	X	X
35	Holm et al., 2010	BPTB	HT	35.9 ± .0	57	33	24	10.5	0.9	Short	X	X	✓	✓	X	X	X
36	Horstmann et al., 2022	HT	QT	28.4 ± 8.0	51	33	18	24	2	Medium	✓	✓	X	✓	X	X	X
37	Ibrahim et al., 2005	BPTB	HT	22.3 ± 8.0	86	86	0	81	6.8	Long	X	✓	X	✓	✓	✓	X
38	Iliopoulos et al., 2017	BPTB	HT	25.8 ± 9.6	20	20	0	9	0.8	Short	✓	✓	X	X	✓	✓	X
41	Jansson et al., 2003	BPTB	HT	.0 ± 3.9	89	0	0	24	2	Medium	X	✓	✓	✓	✓	✓	X
43	Kautzner et al., 2015	BPTB	HT	26.0 ± 6.1	147	0	147	24	2	Medium	X	✓	X	✓	X	X	X
44	Komzak et al., 2022	BPTB	QT	30.4 ± .0	72	44	36	28	2.3	Medium	✓	✓	X	X	X	X	X
45	Konrads et al., 2016	BPTB	HT	29.8 ± 6.3	62	45	17	120	10	Long	✓	✓	✓	✓	✓	✓	X
46	Laoruengthana et al., 2009	BPTB	HT	29.3 ± 9.4	28	0	0	19	1.6	Medium	✓	✓	X	✓	✓	X	X
47	Lawhorn et al., 2012	HT	TA	32.5 ± 6.3	92	70	32	24	2	Medium	✓	X	X	✓	✓	X	X
49	Laxdal et al., 2005	BPTB	HT	26.5 ± 8.9	118	0	0	26	2.2	Medium	✓	✓	✓	✓	X	X	X
51	Li et al., 2015	HT	TA	30.2 ± 8.3	64	32	32	71.4	6	Long	✓	✓	✓	✓	✓	✓	X
52	Lidén et al., 2007	BPTB	HT	28.5 ± 6.8	71	49	22	87.1	7.3	Long	✓	✓	✓	✓	X	✓	X
53	Lind et al., 2020	HT	QT	27.2 ± 7.0	96	54	45	24	2	Medium	✓	X	✓	✓	✓	X	X
54	Lucidi et al., 2025	BPTB	HT	53.7 ± 9.4	39	0	0	240	20	Long	✓	✓	✓	✓	✓	X	✓
55	Lui et al., 2012	BPTB	HT	25.5 ± 11.9	55	55	0	12	1	Short	✓	✓	X	✓	X	✓	X
56	Lund et al., 2014	BPTB	QT	30.5 ± 14.7	39	32	7	24	2	Medium	✓	X	X	✓	X	X	X
58	Malik et al., 2026	HT	PLT	28.0 ± 4.7	50	35	15	6	0.5	Short	✓	✓	✓	✓	✓	✓	X
62	Matsumoto et al., 2006	BPTB	HT	24.1 ± 8.5	72	41	31	83.9	7	Long	✓	X	X	✓	X	X	X
66	Mohtadi et al., 2016	BPTB	HT	28.5 ± 6.5	330	179	151	24	2	Medium	X	X	X	✓	X	✓	✓
68	Moraiti et al., 2010	BPTB	HT	23.5 ± 9.6	12	12	0	23	2	Medium	✓	✓	✓	✓	X	✓	✓
70	Noh et al., 2011	HT	AT	25.7 ± 9.8	65	56	9	29.9	2.5	Medium	✓	✓	✓	✓	✓	✓	✓
73	Papalia et al., 2015	BPTB	HT	25.1 ± 2.5	40	0	40	6	0.5	Short	✓	✓	X	X	X	X	X
76	Pinczewski et al., 2007	BPTB	HT	24.5 ± 7.9	180	95	85	120	10	Long	✓	✓	X	✓	✓	✓	X
77	Popovic et al., 2024	BPTB	HT	27.0 ± .4	144	0	0	228	19	Long	X	✓	✓	✓	✓	✓	X
82	Sajovic et al., 2018	BPTB	HT	44.0 ± 6.8	48	28	20	204	17	Long	✓	✓	X	✓	✓	✓	X
84	Sinding et al., 2020	HT	QT	28.5 ± 6.5	85	48	37	12.5	1	Medium	✓	X	X	X	X	X	X
85	Smith et al., 2020	BPTB	HT	17.7 ± 8.2	64	28	36	24	2	Medium	✓	X	X	✓	X	X	✓
87	Sonnery-Cottet et al., 2026	BPTB	HT	25.0 ± 6.3	556	447	109	62.7	5.2	Long	✓	✓	✓	✓	✓	X	✓
89	Tang et al., 2023	HT	QT	28.2 ± 2.4	33	30	3	24	2	Medium	✓	X	X	X	X	X	X
90	Taylor et al., 2009	BPTB	HT	21.9 ± 4.5	64	53	11	35.4	3	Medium	✓	✓	✓	✓	✓	X	X
93	Vilchez-Cavazos et al., 2020	HT	QT	24.5 ± 7.3	28	23	6	12	1	Short	✓	✓	✓	✓	X	X	X
95	Webster et al., 2001	BPTB	HT	26.5 ± 5.6	61	43	18	24	2	Medium	✓	X	X	✓	X	X	X
96	Wipfler et al., 2011	BPTB	HT	32.1 ± 8.5	62	37	25	108	9	Long	✓	✓	✓	✓	✓	✓	✓
97	Yadav et al., 2025	HT	QT	29.4 ± 6.2	69	62	7	12	1	Short	X	✓	X	✓	✓	✓	X
98	Yoo et al., 2017	HT	TA	27.0 ± 8.6	132	120	12	33.7	2.8	Medium	X	✓	✓	✓	✓	✓	X
100	Zaffagnini et al., 2011	BPTB	HT	26.5 ± 8.8	79	42	37	103.2	8.6	Long	✓	X	X	✓	X	X	X

ID	Study	Arms		Covariates				Follow-up duration	
		Intervention 1	Intervention 2	Age, <i>Mean ± SD</i>	Sample size, <i>n</i>			Months	Years
					Total, <i>n</i>	Male, <i>n</i>	Female, <i>n</i>		
1	Ageberg et al., 2009	BPTB	HT	30 ± 5.1	36	27	9	36	3
2	Aglietti et al., 2004	BPTB	HT	25 ± 5.1	120	92	28	24	2
3	Ahldén et al., 2009	BPTB	HT	28 ± 5.1	47	32	15	88.5	7.4
5	Anderson et al., 2001	BPTB	HT	20.6 ± 5.9	103	68	35	35.9	3
6	Anz et al., 2023	BPTB	HT	28.5 ± 6.4	20	7	13	12	1
7	Aune et al., 2001	BPTB	HT	26 ± 5.8	72	40	32	24	2
9	Barenius et al., 2010	BPTB	HT	34 ± 7.9	153	89	64	96	8
10	Barié et al., 2020	BPTB	QT	30.6 ± 8.1	60	34	26	123	10.3
11	Beynnon et al., 2002	BPTB	HT	29.2 ± 7	56	31	25	36	3
12	Bi et al., 2018	HT	PLT	28.5 ± 7.6	124	65	59	30	2.5
13	Björnsson et al., 2016	BPTB	HT	27.5 ± 6.3	147	95	52	197.3	16.5
14	Bottoni et al., 2015	HT	TA	29.1 ± 6.6	97	84	13	120	10
18	Butt et al., 2024	HT	PLT	28.5 ± 8.3	60	59	1	60	5
19	Calvert et al., 2025	HT	QT	28.8 ± 5.6	112	56	56	24	2
23	Ejerhed et al., 2003	BPTB	HT	29 ± 4.6	66	46	20	24	2
25	Eriksson et al., 2001	BPTB	HT	27 ± 7.9	0	0	0	6.7	0.6
26	Feller & Webster, 2003	BPTB	HT	26.1 ± 9.7	65	47	18	36	3
30	Gifstad et al., 2013	BPTB	HT	27 ± 6	0	0	0	84	7
32	Guglielmetti et al., 2021	BPTB	HT	24.9 ± 6	62	42	20	24	2
33	Harilainen et al., 2006	BPTB	HT					60.6	5.1
34	Heijne & Werner, 2010	BPTB	HT	29.5 ± 5.4	68	36	32	24.7	2.1
35	Holm et al., 2010	BPTB	HT	35.9 ± 0	57	33	24	10.5	0.9
36	Horstmann et al., 2022	HT	QT	28.4 ± 8	51	33	18	24	2
37	Ibrahim et al., 2005	BPTB	HT	22.3 ± 8	86	86	0	81	6.8
38	Iliopoulos et al., 2017	BPTB	HT	25.8 ± 9.6	20	20	0	9	0.8
41	Jansson et al., 2003	BPTB	HT					24	2
43	Kautzner et al., 2015	BPTB	HT	26 ± 6.1	147	0	147	24	2
44	Komzak et al., 2022	BPTB	QT	30.4 ± 0	80	44	36	28	2.3
45	Konrads et al., 2016	BPTB	HT	29.8 ± 6.3	62	45	17	120	10
46	Laoruengthana et al., 2009	BPTB	HT	29.3 ± 9.4	0	0	0	19	1.6
47	Lawhorn et al., 2012	HT	TA	32.5 ± 6.3	102	70	32	24	2
49	Laxdal et al., 2005	BPTB	HT	26.5 ± 8.9	0	0	0	26	2.2
51	Li et al., 2015	HT	TA	30.2 ± 8.3	64	32	32	71.4	6
52	Lidén et al., 2007	BPTB	HT	28.5 ± 6.8	71	49	22	87.1	7.3
53	Lind et al., 2020	HT	QT	27.2 ± 7	99	54	45	24	2
54	Lucidi et al., 2025	BPTB	HT	53.7 ± 9.4				240	20
55	Lui et al., 2012	BPTB	HT	25.5 ± 11.9	55	55	0	12	1
56	Lund et al., 2014	BPTB	QT	30.5 ± 14.7	39	32	7	24	2
58	Malik et al., 2026	HT	PLT	28 ± 4.7	50	35	15	6	0.5
62	Matsumoto et al., 2006	BPTB	HT	24.1 ± 8.5	72	41	31	83.9	7
66	Mohtadi et al., 2016	BPTB	HT	28.5 ± 6.5	330	179	151	24	2
68	Moraiti et al., 2010	BPTB	HT	23.5 ± 9.6	12	12	0	23	2
70	Noh et al., 2011	HT	AT	25.7 ± 9.8	65	56	9	29.9	2.5
73	Papalia et al., 2015	BPTB	HT	25.1 ± 2.5	40	0	40	6	0.5
76	Pinczewski et al., 2007	BPTB	HT	24.5 ± 7.9	180	95	85	120	10
77	Popovic et al., 2024	BPTB	HT	27 ± 0.4	0	0	0	228	19
82	Sajovic et al., 2018	BPTB	HT	44 ± 6.8	48	28	20	204	17
84	Sinding et al., 2020	HT	QT	28.5 ± 6.5	85	48	37	12.5	1
85	Smith et al., 2020	BPTB	HT	17.7 ± 8.2	64	28	36	24	2
87	Sonnery-Cottet et al., 2026	BPTB	HT	25 ± 6.3	556	447	109	62.7	5.2
89	Tang et al., 2023	HT	QT	28.2 ± 2.4	33	30	3	24	2

90	Taylor et al., 2009	BPTB	HT	21.9 ± 4.5	64	53	11	35.4	3
93	Vilchez-Cavazos et al., 2020	HT	QT	24.5 ± 7.3	29	23	6	12	1
95	Webster et al., 2001	BPTB	HT	26.5 ± 5.6	61	43	18	24	2
96	Wipfler et al., 2011	BPTB	HT	32.1 ± 8.5	62	37	25	108	9
97	Yadav et al., 2025	HT	QT	29.4 ± 6.2	69	62	7	12	1
98	Yoo et al., 2017	HT	TA	27 ± 8.6	132	120	12	33.7	2.8
100	Zaffagnini et al., 2011	BPTB	HT	26.5 ± 8.8	79	42	37	103.2	8.6

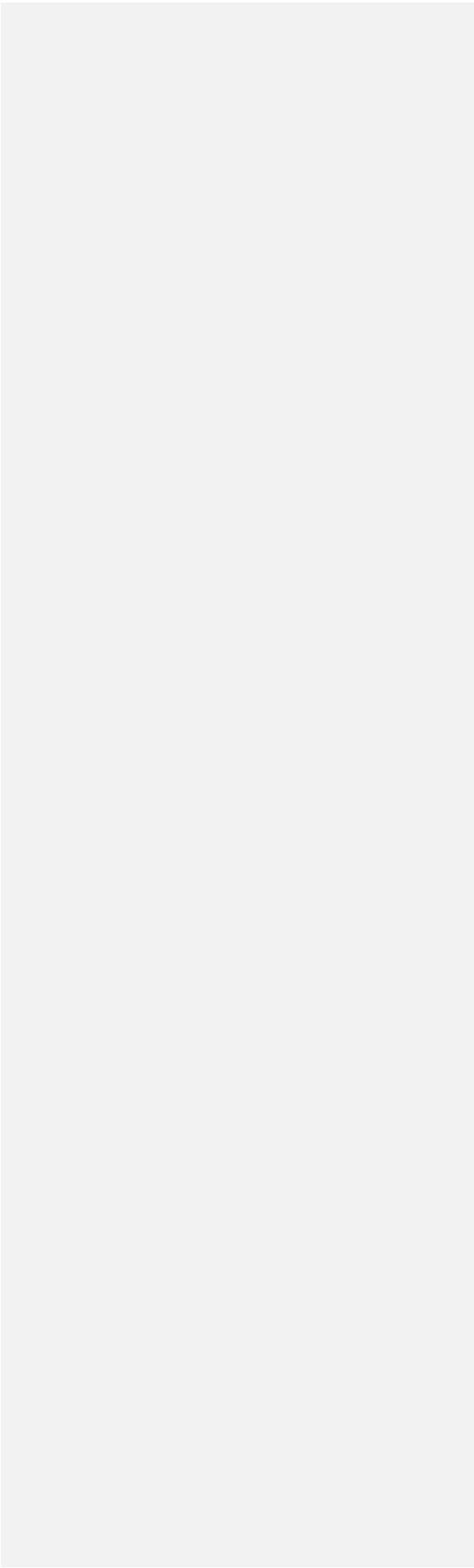
Supplementary Table 1 Eligibility crteria for inclusion and exclusion of studies.

	Inclusion criteria	Exclusion criteria
<i>Population/participants</i>	Adult patients (mean age of all patients ≥ 18 years).	Studies pediatric and adolescent population.
<i>Interventions</i>	Primary anterior cruciate ligament reconstruction surgery	Revision, repair and lateral extra-articular tenodesis (LET) anterior cruciate ligament surgeries.
<i>Comparators</i>	Bone-patellar tendon-bone, hamstring tendon, quadriceps tendon, and peroneus longus tendon autologous grafts. Achilles tendon and tibialis anterior tendon allogenic grafts.	All other autologous and allogenic grafts.
<i>Outcomes</i>	IKDC subjective knee form, Lysholm knee scoring scale, and Tegner activity scale, Instrumented laxity, Pivot-shift, and Lachman test, graft re-rupture, clinical failure (pivot-shift and Lachman ≥ grade 2+ and 3+), and revision surgery.	All other outcome measures (e.g., Marx activity scale, KOOS (subscales: QoL, ADL, Sports & Recreation, Symptoms, and Pain), Anterior drawer test, muscle strength recovery (Q-LSI and H-LSI).
<i>Study designs</i>	Randomised controlled trials (RCTs)	Observational studies, case series and reports.

Supplementary Table 2 Database search strategies

Data source	Full search strategy
PubMed/MEDLINE	((("anterior cruciate ligament"[mh] OR "anterior cruciate ligament"[tiab] OR "anterior cruciate ligament reconstruction"[tiab] OR "acl"[tiab])) AND (("bone-patellar tendon-bone"[tiab] OR "bone patellar tendon bone"[tiab] OR "patellar tendon"[tiab] OR "bptb"[tiab]) OR ("hamstring"[tiab] OR "hamstring tendon"[tiab] OR "semitendinosus"[tiab] OR "gracilis"[tiab]) OR ("quadriceps"[tiab] OR "quadriceps tendon"[tiab] OR "qt"[tiab]) OR ("peroneus"[tiab] OR "peroneus longus"[tiab] OR "fibularis longus"[tiab] OR ("achilles"[tiab] OR "achilles tendon"[tiab]) OR ("tibialis"[tiab] OR "tibialis anterior"[tiab] OR "tibialis posterior"[tiab])) AND (("randomized controlled trial"[pt] OR "randomized controlled trial"[tiab] OR "randomised controlled trial"[tiab]) NOT ("review"[pt] OR "review"[tiab] OR "systematic review"[pt] OR "systematic review"[tiab] OR "meta-analysis"[pt] OR "meta-analysis"[tiab]))
Embase	((('anterior cruciate ligament'/exp OR 'anterior cruciate ligament':ti,ab OR 'anterior cruciate ligament reconstruction':ti,ab OR 'acl':ti,ab)) AND (('bone-patellar tendon-bone':ti,ab OR 'bone patellar tendon bone':ti,ab OR 'patellar tendon':ti,ab OR 'bptb':ti,ab) OR ('hamstring':ti,ab OR 'hamstring tendon':ti,ab OR 'semitendinosus':ti,ab OR 'gracilis':ti,ab) OR ('quadriceps':ti,ab OR 'quadriceps tendon':ti,ab OR 'qt':ti,ab) OR ('peroneus':ti,ab OR 'peroneus longus':ti,ab OR 'fibularis longus':ti,ab) OR ('achilles':ti,ab OR 'achilles tendon':ti,ab) OR ('tibialis':ti,ab OR 'tibialis anterior':ti,ab OR 'tibialis posterior':ti,ab)) AND (('randomized controlled trial':it,ti,ab OR 'randomized controlled trial':ti,ab OR 'randomised controlled trial':ti,ab) NOT ('review':it,ti,ab OR 'review':ti,ab OR 'systematic review':it,ti,ab OR 'systematic review':ti,ab OR 'meta-analysis':it,ti,ab OR 'meta-analysis':ti,ab))
Web of Science Core Collection	((((TMIC=(anterior cruciate ligament)) OR (TI=(anterior cruciate ligament) OR AB=(anterior cruciate ligament)) OR (TI=(anterior cruciate ligament reconstruction) OR AB=(anterior cruciate ligament reconstruction)) OR (TI=(acl) OR AB=(acl)))) AND (((TI=(bone-patellar tendon-bone) OR AB=(bone-patellar tendon-bone)) OR (TI=(bone patellar tendon bone) OR AB=(bone patellar tendon bone)) OR (TI=(patellar tendon) OR AB=(patellar tendon)) OR (TI=(bptb) OR AB=(bptb)))) OR ((TI={hamstring} OR AB={hamstring}) OR (TI={hamstring tendon} OR AB={hamstring tendon})) OR (TI={semitendinosus} OR AB={semitendinosus}) OR (TI={gracilis} OR AB={gracilis})) OR ((TI={quadriceps} OR AB={quadriceps}) OR (TI={quadriceps tendon} OR AB={quadriceps tendon})) OR (TI={qt} OR AB={qt})) OR ((TI={peroneus} OR AB={peroneus}) OR (TI={peroneus longus} OR AB={peroneus longus}) OR (TI={fibularis longus} OR AB={fibularis longus})) OR ((TI={achilles} OR AB={achilles}) OR (TI={achilles tendon} OR AB={achilles tendon})) OR ((TI={tibialis} OR AB={tibialis}) OR (TI={tibialis anterior} OR AB={tibialis anterior}) OR (TI={tibialis posterior} OR AB={tibialis posterior}))) AND (((TS=(randomized controlled trial)) OR (TI=(randomized controlled trial) OR AB=(randomized controlled trial)) OR (TI=(randomised controlled trial) OR AB=(randomised controlled trial))) NOT ((TS=(review)) OR (TI=(review) OR AB=(review)) OR (TS=(systematic review)) OR (TI=(systematic review) OR AB=(systematic review)) OR (TS=(meta-analysis)) OR (TI=(meta-analysis) OR AB=(meta-analysis))))
Cochrane Central Register of Controlled Trials (CENTRAL)	([mh] ("anterior cruciate ligament") OR "anterior cruciate ligament":ti,ab OR "anterior cruciate ligament reconstruction":ti,ab OR "acl":ti,ab)) AND (("bone-patellar tendon-bone":ti,ab OR "bone patellar tendon bone":ti,ab OR "patellar tendon":ti,ab OR "bptb":ti,ab) OR ("hamstring":ti,ab OR "hamstring tendon":ti,ab OR "semitendinosus":ti,ab OR "gracilis":ti,ab) OR ("quadriceps":ti,ab OR "quadriceps tendon":ti,ab OR "qt":ti,ab) OR ("peroneus":ti,ab OR "peroneus longus":ti,ab OR "fibularis longus":ti,ab) OR ("achilles":ti,ab OR "achilles tendon":ti,ab) OR ("tibialis":ti,ab OR "tibialis anterior":ti,ab OR "tibialis posterior":ti,ab)) AND (("randomized controlled trial":pt OR "randomized controlled trial":ti,ab OR "randomised controlled trial":ti,ab) NOT ("review":pt OR

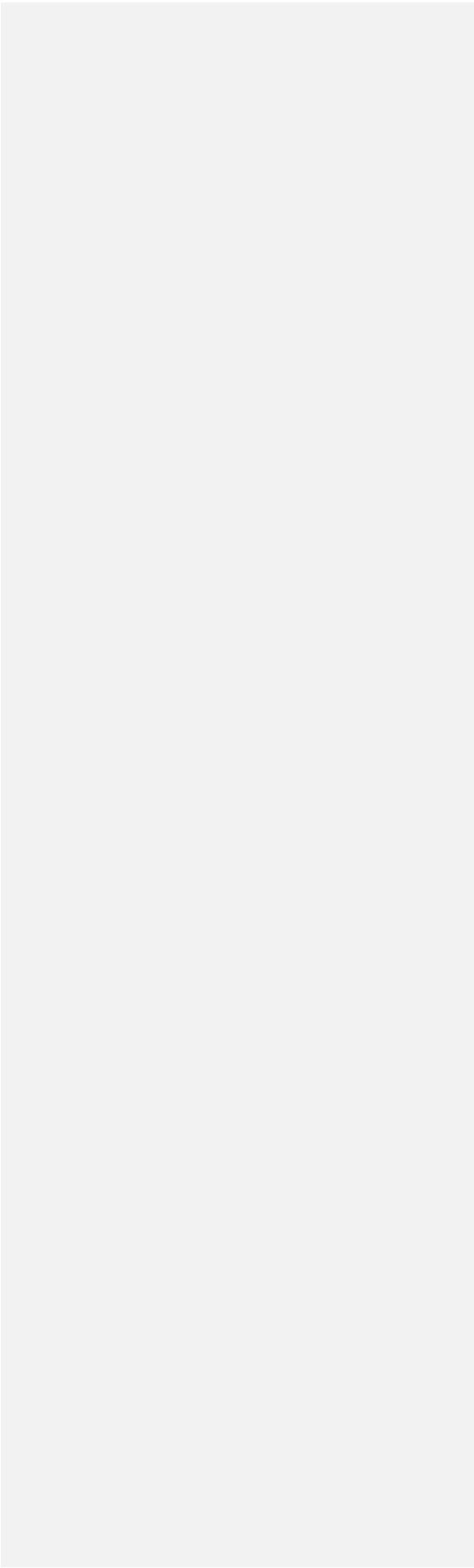
"review":ti,ab OR "systematic review":pt OR "systematic review":ti,ab OR "meta-analysis":pt OR "meta-analysis":ti,ab))



Supplementary Table 3 IKDC subjective knee form (IKDC-SKF) score

ID	Study	Subgroup	Sample size	Mean ± SD
10	Barié et al., 2020	BPTB	43	91.00 ± 7.30
10	Barié et al., 2020	QT	43	92.00 ± 11.50
12	Bi et al., 2018	HT	62	90.40 ± 7.10
12	Bi et al., 2018	PLT	62	89.30 ± 8.40
13	Björnsson et al., 2016	BPTB	61	67.30 ± 20.80
13	Björnsson et al., 2016	HT	86	74.00 ± 18.80
14	Bottoni et al., 2015	HT	44	79.50 ± 24.70
14	Bottoni et al., 2015	TA	36	81.10 ± 23.10
18	Butt et al., 2024	HT	30	89.70 ± 5.70
18	Butt et al., 2024	PLT	30	89.90 ± 9.80
19	Calvert et al., 2025	HT	92	92.00 ± 8.00
19	Calvert et al., 2025	QT	89	89.00 ± 10.00
32	Guglielmetti et al., 2021	BPTB	31	88.75 ± 9.27
32	Guglielmetti et al., 2021	HT	31	91.00 ± 9.07
36	Horstmann et al., 2022	HT	26	83.70 ± 12.70
36	Horstmann et al., 2022	QT	23	89.30 ± 12.20
38	Iliopoulos et al., 2017	BPTB	45	83.20 ± 2.80
38	Iliopoulos et al., 2017	HT	45	82.78 ± 2.80
44	Komzak et al., 2022	BPTB	36	74.20 ± 10.20
44	Komzak et al., 2022	QT	36	78.30 ± 10.50
51	Li et al., 2015	HT	32	87.50 ± 3.20
51	Li et al., 2015	TA	32	83.80 ± 6.90
53	Lind et al., 2020	HT	49	78.00 ± 18.00
53	Lind et al., 2020	QT	47	82.00 ± 14.00
54	Lucidi et al., 2025	BPTB	19	78.40 ± 13.70
54	Lucidi et al., 2025	HT	20	81.60 ± 11.90
55	Lui et al., 2012	BPTB	14	71.90 ± 15.20
55	Lui et al., 2012	HT	17	73.10 ± 12.30
56	Lund et al., 2014	BPTB	18	75.00 ± 13.00
56	Lund et al., 2014	QT	21	76.00 ± 16.00
58	Malik et al., 2026	HT	25	78.90 ± 7.40
58	Malik et al., 2026	PLT	25	82.60 ± 6.30
62	Matsumoto et al., 2006	BPTB	37	84.30 ± 11.00
62	Matsumoto et al., 2006	HT	35	86.80 ± 8.30
68	Moraiti et al., 2010	BPTB	6	92.00 ± 7.00
68	Moraiti et al., 2010	HT	6	90.00 ± 10.00
73	Papalia et al., 2015	BPTB	20	86.90 ± 11.70
73	Papalia et al., 2015	HT	20	90.97 ± 5.10
84	Sinding et al., 2020	HT	43	76.00 ± 15.00
84	Sinding et al., 2020	QT	42	76.00 ± 17.00
85	Smith et al., 2020	BPTB	32	89.40 ± 9.20
85	Smith et al., 2020	HT	32	94.80 ± 6.40
87	Sonnery-cottet et al., 2026	BPTB	296	89.40 ± 10.90
87	Sonnery-cottet et al., 2026	HT	297	91.00 ± 10.30
89	Tang et al., 2023	HT	16	90.13 ± 6.99

89	Tang et al, 2023	QT	17	87.12 ± 6.61
93	Vilchez-Cavazos et al., 2020	BPTB	14	90.00 ± 1.76
93	Vilchez-Cavazos et al., 2020	QT	14	88.17 ± 2.49
96	Wipfler et al., 2011	BPTB	29	86.72 ± 2.27
96	Wipfler et al., 2011	HT	25	92.95 ± 1.99
100	Zaffagnini et al., 2011	BPTB	40	82.00 ± 20.00
100	Zaffagnini et al., 2011	HT	39	88.00 ± 9.00



Supplementary Table 4 Lysholm

ID	Study	Subgroup	Sample size	Mean ± SD
3	Ahldén et al., 2009	BPTB	25	80.70 ± 21.64
3	Ahldén et al., 2009	HT	22	79.16 ± 13.35
9	Barenius et al., 2010	BPTB	78	85.82 ± 11.83
9	Barenius et al., 2010	HT	75	86.42 ± 12.53
10	Barié et al., 2020	BPTB	43	95.20 ± 6.60
10	Barié et al., 2020	QT	43	95.60 ± 7.80
13	Björnsson et al., 2016	BPTB	61	79.40 ± 16.90
13	Björnsson et al., 2016	HT	86	80.70 ± 15.30
18	Butt et al., 2024	HT	30	95.10 ± 5.20
18	Butt et al., 2024	PLT	30	94.50 ± 6.40
19	Calvert et al., 2025	HT	92	94.00 ± 7.00
19	Calvert et al., 2025	QT	89	93.00 ± 8.00
23	Ejerhed et al., 2003	BPTB	32	89.96 ± 13.06
23	Ejerhed et al., 2003	HT	34	86.79 ± 11.71
32	Guglielmetti et al., 2021	BPTB	31	89.87 ± 9.17
32	Guglielmetti et al., 2021	HT	31	91.26 ± 9.36
33	Harilainen et al., 2006	BPTB	37	89.11 ± 15.53
33	Harilainen et al., 2006	HT	37	95.00 ± 15.53
35	Holm et al., 2010	BPTB	28	84.20 ± 15.40
35	Holm et al., 2010	HT	29	86.10 ± 15.10
36	Horstmann et al., 2022	HT	26	83.50 ± 17.40
36	Horstmann et al., 2022	QT	23	90.40 ± 11.90
37	Ibrahim et al., 2005	BPTB	40	91.60 ± 3.94
37	Ibrahim et al., 2005	HT	46	92.70 ± 3.62
38	Iliopoulos et al., 2017	BPTB	45	95.80 ± 2.40
38	Iliopoulos et al., 2017	HT	45	94.00 ± 3.60
43	Kautzner et al., 2015	BPTB	74	88.00 ± 7.50
43	Kautzner et al., 2015	HT	73	90.00 ± 7.60
44	Komzak et al., 2022	BPTB	36	93.40 ± 9.60
44	Komzak et al., 2022	QT	36	90.10 ± 6.70
45	Konrads et al., 2016	BPTB	24	92.00 ± 8.99
45	Konrads et al., 2016	HT	23	91.80 ± 9.33
49	Laxdal et al., 2005	BPTB	40	94.00 ± 18.09
49	Laxdal et al., 2005	HT	78	92.00 ± 13.40
51	Li et al., 2015	HT	32	91.30 ± 11.50
51	Li et al., 2015	TA	32	88.70 ± 8.60
52	Lidén et al., 2007	BPTB	30	81.00 ± 18.38
52	Lidén et al., 2007	HT	32	90.00 ± 12.10
54	Lucidi et al., 2025	BPTB	19	84.80 ± 10.30
54	Lucidi et al., 2025	HT	20	88.60 ± 8.30
55	Lui et al., 2012	BPTB	14	84.00 ± 13.47
55	Lui et al., 2012	HT	17	80.00 ± 15.59

58	Malik et al., 2026	HT	25	85.10 ± 6.40
58	Malik et al., 2026	PLT	25	89.40 ± 5.80
68	Moraiti et al., 2010	BPTB	6	96.00 ± 3.00
68	Moraiti et al., 2010	HT	6	95.00 ± 5.00
70	Noh et al., 2011	HT	33	98.00 ± 3.61
70	Noh et al., 2011	AT	32	99.00 ± 3.63
73	Papalia et al., 2015	BPTB	20	87.40 ± 13.80
73	Papalia et al., 2015	HT	20	93.17 ± 10.30
77	Popovic et al., 2024	BPTB	38	82.00 ± 15.45
77	Popovic et al., 2024	HT	33	85.00 ± 14.67
87	Sonnery-Cottet et al., 2026	BPTB	296	89.00 ± 12.40
87	Sonnery-Cottet et al., 2026	HT	297	91.10 ± 11.90
89	Tang et al., 2023	HT	16	90.06 ± 7.33
89	Tang et al., 2023	QT	17	90.00 ± 4.26
90	Taylor et al., 2009	BPTB	21	90.30 ± 12.19
90	Taylor et al., 2009	HT	23	90.40 ± 12.75
93	Vilchez-Cavazos et al., 2020	BPTB	14	96.00 ± 2.93
93	Vilchez-Cavazos et al., 2020	QT	14	95.33 ± 2.64
96	Wipfler et al., 2011	BPTB	29	87.28 ± 1.76
96	Wipfler et al., 2011	HT	25	91.82 ± 1.71
97	Yadav et al., 2025	HT	36	92.40 ± .00
97	Yadav et al., 2025	QT	33	88.20 ± .00
98	Yoo et al., 2017	HT	68	96.00 ± 6.99
98	Yoo et al., 2017	TA	64	93.00 ± 5.78

Supplementary Table 5 Tegner Activity Scale

ID	Study	Subgroup	Sample size	Mean ± SD
1	Ageberg et al., 2009	BPTB	16	4.17 ± 1.41
1	Ageberg et al., 2009	HT	20	3.15 ± 1.34
3	Ahldén et al., 2009	BPTB	25	4.87 ± 2.29
3	Ahldén et al., 2009	HT	22	4.86 ± 1.31
9	Barenius et al., 2010	BPTB	78	5.80 ± 1.87
9	Barenius et al., 2010	HT	75	4.07 ± 1.88
10	Barié et al., 2020	BPTB	43	6.10 ± 1.14
10	Barié et al., 2020	QT	43	6.10 ± 1.14
13	Björnsson et al., 2016	BPTB	61	4.10 ± 1.70
13	Björnsson et al., 2016	HT	86	4.00 ± 1.70
14	Bottoni et al., 2015	HT	44	5.10 ± 2.20
14	Bottoni et al., 2015	TA	36	5.10 ± 2.00
23	Ejerhed et al., 2003	BPTB	32	5.77 ± 1.94
23	Ejerhed et al., 2003	HT	34	6.39 ± 1.43
33	Harilainen et al., 2006	BPTB	37	5.89 ± 2.12
33	Harilainen et al., 2006	HT	37	6.00 ± 1.41
34	Heijne & Wemer, 2010	BPTB	30	6.64 ± 2.21
34	Heijne & Wemer, 2010	HT	29	5.00 ± 1.48
41	Jansson et al., 2003	BPTB	43	5.98 ± 1.60
41	Jansson et al., 2003	HT	46	5.82 ± 2.26
45	Konrads et al., 2016	BPTB	24	5.90 ± 1.28
45	Konrads et al., 2016	HT	23	5.10 ± 1.04
49	Laxdal et al., 2005	BPTB	40	6.00 ± 1.85
49	Laxdal et al., 2005	HT	78	4.00 ± 1.64
51	Li et al., 2015	HT	32	7.30 ± 1.30
51	Li et al., 2015	TA	32	7.00 ± 2.10
52	Lidén et al., 2007	BPTB	30	5.00 ± 1.72
52	Lidén et al., 2007	HT	32	6.00 ± 1.69
53	Lind et al., 2020	HT	49	5.90 ± 1.30
53	Lind et al., 2020	QT	47	6.00 ± 1.50
54	Lucidi et al., 2025	BPTB	19	3.00 ± 1.08
54	Lucidi et al., 2025	HT	20	4.00 ± .80
58	Malik et al., 2026	HT	25	5.10 ± 1.00
58	Malik et al., 2026	PLT	25	5.60 ± .90
70	Noh et al., 2011	HT	33	6.00 ± .96
70	Noh et al., 2011	AT	32	6.00 ± 1.21
77	Popovic et al., 2024	BPTB	38	4.00 ± 1.64
77	Popovic et al., 2024	HT	33	5.00 ± 2.16
87	Sonnery-Cottet et al., 2026	BPTB	296	6.60 ± 1.60
87	Sonnery-Cottet et al., 2026	HT	297	6.70 ± 1.70
90	Taylor et al., 2009	BPTB	20	6.80 ± 3.27
90	Taylor et al., 2009	HT	24	5.30 ± 3.58

96	Wipfler et al., 2011	BPTB	29	6.20 ± .35
96	Wipfler et al., 2011	HT	25	6.14 ± .37
98	Yoo et al., 2017	HT	68	5.00 ± 1.48
98	Yoo et al., 2017	TA	64	5.00 ± 1.07

Supplementary Table 6 Side-to-side difference in instrumented laxity

ID	Study	Subgroup	Sample size	Mean ± SD
2	Aglietti et al., 2004	BPTB	60	2.20 ± 1.08
2	Aglietti et al., 2004	HT	60	1.95 ± 1.30
3	Ahldén et al., 2009	BPTB	25	2.60 ± 3.30
3	Ahldén et al., 2009	HT	22	1.40 ± 2.60
5	Anderson et al., 2001	BPTB	35	2.10 ± 2.00
5	Anderson et al., 2001	HT	33	3.10 ± 2.30
7	Aune et al., 2001	BPTB	32	3.30 ± 2.60
7	Aune et al., 2001	HT	29	2.90 ± 2.40
10	Barié et al., 2020	BPTB	43	1.05 ± 1.36
10	Barié et al., 2020	QT	43	1.00 ± 1.10
12	Bi et al., 2018	HT	62	1.71 ± .57
12	Bi et al., 2018	PLT	62	1.85 ± .77
23	Ejerhed et al., 2003	BPTB	32	2.29 ± 3.99
23	Ejerhed et al., 2003	HT	34	2.47 ± 3.47
25	Eriksson et al., 2001	BPTB	42	0.95 ± .91
25	Eriksson et al., 2001	HT	47	1.10 ± .95
26	Feller & Webster, 2003	BPTB	27	0.50 ± 1.00
26	Feller & Webster, 2003	HT	21	1.10 ± 1.20
30	Gifstad et al., 2013	BPTB	41	1.40 ± 1.80
30	Gifstad et al., 2013	HT	36	1.40 ± 1.40
33	Harilainen et al., 2006	BPTB	37	1.50 ± 3.00
33	Harilainen et al., 2006	HT	37	1.20 ± 3.90
35	Holm et al., 2010	BPTB	28	3.00 ± 3.20
35	Holm et al., 2010	HT	29	2.00 ± 3.50
36	Horstmann et al., 2022	HT	26	0.20 ± 2.20
36	Horstmann et al., 2022	QT	23	0.70 ± 1.10
41	Jansson et al., 2003	BPTB	43	1.70 ± 2.63
41	Jansson et al., 2003	HT	46	1.20 ± 2.65
43	Kautzner et al., 2015	BPTB	74	0.60 ± .21
43	Kautzner et al., 2015	HT	73	0.50 ± .42
45	Konrads et al., 2016	BPTB	24	1.80 ± .51
45	Konrads et al., 2016	HT	23	2.20 ± .78
46	Laoruengthana et al., 2009	BPTB	15	1.22 ± .69
46	Laoruengthana et al., 2009	HT	13	0.96 ± .42
49	Laxdal et al., 2005	BPTB	40	1.00 ± 3.01
49	Laxdal et al., 2005	HT	78	1.50 ± 3.12
51	Li et al., 2015	HT	32	2.10 ± 1.60
51	Li et al., 2015	TA	32	3.50 ± 1.20
52	Lidén et al., 2007	BPTB	30	2.30 ± 3.06
52	Lidén et al., 2007	HT	32	2.70 ± 3.39
53	Lind et al., 2020	HT	49	1.90 ± 1.70
53	Lind et al., 2020	QT	47	1.80 ± 1.00
54	Lucidi et al., 2025	BPTB	19	-0.10 ± 1.60
54	Lucidi et al., 2025	HT	20	1.50 ± 1.20
55	Lui et al., 2012	BPTB	14	3.70 ± 1.40
55	Lui et al., 2012	HT	17	3.10 ± 1.60
56	Lund et al., 2014	BPTB	18	1.10 ± 1.40

56	Lund et al., 2014	QT	21	0.80 ± 1.70
58	Malik et al., 2026	HT	25	2.10 ± 1.00
58	Malik et al., 2026	PLT	25	1.80 ± .90
62	Matsumoto et al., 2006	BPTB	37	1.20 ± 2.10
62	Matsumoto et al., 2006	HT	35	1.70 ± 1.40
76	Pinczewski et al., 2007	BPTB	54	1.20 ± 1.69
76	Pinczewski et al., 2007	HT	58	1.60 ± 1.55
77	Popovic et al., 2024	BPTB	38	1.30 ± 2.20
77	Popovic et al., 2024	HT	33	1.50 ± 3.30
85	Smith et al., 2020	BPTB	32	0.00 ± .80
85	Smith et al., 2020	HT	32	0.30 ± .70
87	Sonnery-cottet et al., 2026	BPTB	296	1.20 ± 1.40
87	Sonnery-cottet et al., 2026	HT	297	0.90 ± 1.00
95	Webster et al., 2001	BPTB	23	1.10 ± .65
95	Webster et al., 2001	HT	31	1.70 ± .97
96	Wipfler et al., 2011	BPTB	29	0.90 ± .27
96	Wipfler et al., 2011	HT	25	0.64 ± .36
97	Yadav et al., 2025	HT	36	3.77 ± 2.01
97	Yadav et al., 2025	QT	33	2.73 ± 1.64
100	Zaffagnini et al., 2011	BPTB	40	0.40 ± .60
100	Zaffagnini et al., 2011	HT	39	1.10 ± 1.90

Supplementary Table 7 Pivot-shift test

Commented [dongwkim14]: Fill it later

ID	Study	Subgroup	Sample size	0 (Normal)	1+ (Glide)	2+ (Clunk)	3+ (Gross)	≥ 1+		≥ 2+	
								Event	Proportion	Event	Proportion
2	Aglietti et al., 2004	BPTB	60	50 (83%)	10 (17%)	0 (0%)	0 (0%)	10 (17%)	0.174	0 (0%)	0.008
2	Aglietti et al., 2004	HT	60	49 (82%)	11 (18%)	0 (0%)	0 (0%)	11 (18%)	0.19	0 (0%)	0.008
5	Anderson et al., 2001	BPTB	35	17 (49%)	3 (9%)	8 (23%)	7 (20%)	3 (9%)	0.521	15 (43%)	0.437
5	Anderson et al., 2001	HT	70	34 (49%)	9 (13%)	16 (23%)	11 (16%)	9 (13%)	0.518	27 (39%)	0.390
11	Beynnon et al., 2002	BPTB	22	19 (86%)	3 (14%)	0 (0%)	0 (0%)	3 (14%)	0.156	0 (0%)	0.022
11	Beynnon et al., 2002	HT	22	13 (59%)	3 (14%)	6 (27%)	0 (0%)	3 (14%)	0.422	6 (27%)	0.289
12	Bi et al., 2018	HT	62	58 (94%)	3 (5%)	1 (2%)	0 (0%)	3 (5%)	0.072	1 (2%)	0.024
12	Bi et al., 2018	PLT	62	57 (92%)	5 (8%)	0 (0%)	0 (0%)	5 (8%)	0.088	0 (0%)	0.008
13	Björnsson et al., 2016	BPTB	60	34 (57%)	14 (23%)	12 (20%)	0 (0%)	14 (23%)	0.438	12 (20%)	0.207
13	Björnsson et al., 2016	HT	86	58 (67%)	15 (17%)	12 (14%)	1 (1%)	15 (17%)	0.329	13 (15%)	0.156
25	Eriksson et al., 2001	BPTB	42	36 (86%)	6 (14%)	0 (0%)	0 (0%)	6 (14%)	0.153	0 (0%)	0.012
25	Eriksson et al., 2001	HT	47	28 (60%)	19 (40%)	0 (0%)	0 (0%)	19 (40%)	0.411	0 (0%)	0.011
26	Feller & Webster, 2003	BPTB	42	40 (95%)	2 (5%)	0 (0%)	0 (0%)	2 (5%)	0.059	0 (0%)	0.012
26	Feller & Webster, 2003	HT	47	38 (81%)	9 (19%)	0 (0%)	0 (0%)	9 (19%)	0.2	0 (0%)	0.011
32	Guglielmetti et al., 2021	BPTB	31	24 (77%)	5 (16%)	0 (0%)	2 (6%)	5 (16%)	0.238	2 (6%)	0.079
32	Guglielmetti et al., 2021	HT	31	20 (65%)	8 (26%)	2 (6%)	1 (3%)	8 (26%)	0.365	3 (10%)	0.111
33	Harilainen et al., 2006	BPTB	37	33 (89%)	4 (11%)	0 (0%)	0 (0%)	4 (11%)	0.12	0 (0%)	0.013
33	Harilainen et al., 2006	HT	37	33 (89%)	3 (8%)	1 (3%)	0 (0%)	3 (8%)	0.12	1 (3%)	0.040
34	Heijne & Werner, 2010	BPTB	30	17 (57%)	10 (33%)	3 (10%)	0 (0%)	10 (33%)	0.443	3 (10%)	0.115
34	Heijne & Werner, 2010	HT	29	8 (28%)	13 (45%)	7 (24%)	1 (3%)	13 (45%)	0.729	8 (28%)	0.288
36	Horstmann et al., 2022	HT	26	26 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.019	0 (0%)	0.019
36	Horstmann et al., 2022	QT	23	23 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.021	0 (0%)	0.021
37	Ibrahim et al., 2005	BPTB	40	35 (88%)	5 (13%)	0 (0%)	0 (0%)	5 (13%)	0.136	0 (0%)	0.012
37	Ibrahim et al., 2005	HT	45	37 (82%)	8 (18%)	0 (0%)	0 (0%)	8 (18%)	0.187	0 (0%)	0.011
41	Jansson et al., 2003	BPTB	43	40 (93%)	3 (7%)	0 (0%)	0 (0%)	3 (7%)	0.08	0 (0%)	0.011
41	Jansson et al., 2003	HT	46	43 (93%)	3 (7%)	0 (0%)	0 (0%)	3 (7%)	0.075	0 (0%)	0.011
46	Laouenghana et al., 2009	BPTB	15	13 (87%)	2 (13%)	0 (0%)	0 (0%)	2 (13%)	0.161	0 (0%)	0.032
46	Laouenghana et al., 2009	HT	13	11 (85%)	2 (15%)	0 (0%)	0 (0%)	2 (15%)	0.185	0 (0%)	0.037
47	Lawhom et al., 2012	HT	54	48 (89%)	5 (9%)	0 (0%)	1 (2%)	5 (9%)	0.119	1 (2%)	0.028
47	Lawhom et al., 2012	TA	48	44 (92%)	4 (8%)	0%	0%	4 (8%)	0.093	0 (0%)	0.010
51	Li et al., 2015	HT	32	28 (88%)	4 (13%)	0 (0%)	0 (0%)	4 (13%)	0.138	0 (0%)	0.015
51	Li et al., 2015	TA	32	23 (72%)	8 (25%)	1 (3%)	0 (0%)	8 (25%)	0.292	1 (3%)	0.046
53	Lind et al., 2020	HT	49	38 (78%)	11 (22%)	0 (0%)	0 (0%)	11 (22%)	0.232	0 (0%)	0.010
53	Lind et al., 2020	QT	47	39 (83%)	8 (17%)	0 (0%)	0 (0%)	8 (17%)	0.179	0 (0%)	0.011
70	Noh et al., 2011	HT	33	27 (82%)	5 (15%)	1 (3%)	0 (0%)	5 (15%)	0.194	1 (3%)	0.045
70	Noh et al., 2011	AT	32	21 (66%)	8 (25%)	3 (9%)	0 (0%)	8 (25%)	0.354	3 (9%)	0.108
76	Pinczewski et al., 2007	BPTB	53	48 (91%)	5 (9%)	0 (0%)	0 (0%)	5 (9%)	0.103	0 (0%)	0.009
76	Pinczewski et al., 2007	HT	58	50 (86%)	8 (14%)	0 (0%)	0 (0%)	8 (14%)	0.145	0 (0%)	0.009
82	Sajovic et al., 2018	BPTB	24	19 (79%)	5 (21%)	0 (0%)	0 (0%)	5 (21%)	0.224	0 (0%)	0.020
82	Sajovic et al., 2018	HT	24	18 (75%)	6 (25%)	0 (0%)	0 (0%)	6 (25%)	0.265	0 (0%)	0.020
97	Yadav et al., 2025	HT	36	31 (86%)	4 (11%)	1 (3%)	0 (0%)	4 (11%)	0.151	1 (3%)	0.041
97	Yadav et al., 2025	QT	33	33 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.015	0 (0%)	0.015
98	Yoo et al., 2017	HT	68	42 (62%)	24 (35%)	2 (3%)	0 (0%)	24 (35%)	0.387	2 (3%)	0.036
98	Yoo et al., 2017	TA	64	31 (48%)	30 (47%)	1 (2%)	2 (3%)	30 (47%)	0.519	3 (5%)	0.054

Supplementary Table 8 Lachman test

ID	Study	Subgroup	Sample size	0	1+	2+	3+	≥ 1+		≥ 2+	
				(≥ 2.5 mm)	(3-5 mm)	(5-10 mm)	(> 10 mm)	Event	Proportion	Event	Proportion
11	Beynnon et al., 2002	BPTB	22	10 (45%)	10 (45%)	2 (9%)	0 (0%)	12 (55%)	0.556	2 (9%)	0.111
11	Beynnon et al., 2002	HT	22	3 (14%)	6 (27%)	8 (36%)	5 (23%)	19 (86%)	0.867	13 (59%)	0.600
13	Björnsson et al., 2016	BPTB	61	30 (49%)	24 (39%)	6 (10%)	1 (2%)	31 (51%)	0.512	7 (11%)	0.122
13	Björnsson et al., 2016	HT	86	19 (22%)	46 (53%)	12 (14%)	9 (10%)	67 (78%)	0.780	21 (24%)	0.249
23	Ejerhed et al., 2003	BPTB	32	18 (56%)	12 (38%)	2 (6%)	0 (0%)	14 (44%)	0.446	2 (6%)	0.077
23	Ejerhed et al., 2003	HT	33	16 (48%)	17 (52%)	0 (0%)	0 (0%)	17 (52%)	0.522	0 (0%)	0.015
26	Feller & Webster, 2003	BPTB	42	36 (86%)	6 (14%)	0 (0%)	0 (0%)	6 (14%)	0.153	0 (0%)	0.012
26	Feller & Webster, 2003	HT	47	28 (60%)	19 (40%)	0 (0%)	0 (0%)	19 (40%)	0.411	0 (0%)	0.011
30	Gifstad et al., 2013	BPTB	45	29 (64%)	15 (33%)	1 (2%)	0 (0%)	16 (36%)	0.363	1 (2%)	0.033
30	Gifstad et al., 2013	HT	32	19 (59%)	12 (38%)	1 (3%)	0 (0%)	13 (41%)	0.415	1 (3%)	0.046
32	Guglielmetti et al., 2021	BPTB	31	22 (71%)	7 (23%)	1 (3%)	1 (3%)	9 (29%)	0.302	2 (6%)	0.079
32	Guglielmetti et al., 2021	HT	31	22 (71%)	6 (19%)	2 (6%)	1 (3%)	9 (29%)	0.302	3 (10%)	0.111
33	Harilainen et al., 2006	BPTB	37	29 (78%)	8 (22%)	0 (0%)	0 (0%)	8 (22%)	0.227	0 (0%)	0.013
33	Harilainen et al., 2006	HT	37	31 (84%)	5 (14%)	1 (3%)	0 (0%)	6 (16%)	0.173	1 (3%)	0.040
37	Ibrahim et al., 2005	BPTB	40	35 (88%)	5 (13%)	0 (0%)	0 (0%)	5 (13%)	0.136	0 (0%)	0.012
37	Ibrahim et al., 2005	HT	46	39 (85%)	7 (15%)	0 (0%)	0 (0%)	7 (15%)	0.161	0 (0%)	0.011
41	Jansson et al., 2003	BPTB	43	35 (81%)	8 (19%)	0 (0%)	0 (0%)	8 (19%)	0.195	0 (0%)	0.011
41	Jansson et al., 2003	HT	46	38 (83%)	8 (17%)	0 (0%)	0 (0%)	8 (17%)	0.183	0 (0%)	0.011
45	Konrads et al., 2016	BPTB	24	16 (67%)	8 (33%)	0 (0%)	0 (0%)	8 (33%)	0.347	0 (0%)	0.020
45	Konrads et al., 2016	HT	23	12 (52%)	9 (39%)	1 (4%)	1 (4%)	11 (48%)	0.489	2 (9%)	0.106
51	Li et al., 2015	HT	32	26 (81%)	4 (13%)	2 (6%)	0 (0%)	6 (19%)	0.200	2 (6%)	0.077
51	Li et al., 2015	TA	32	19 (59%)	9 (28%)	3 (9%)	1 (3%)	13 (41%)	0.415	4 (13%)	0.138
52	Lidén et al., 2007	BPTB	30	8 (27%)	21 (70%)	1 (3%)	0 (0%)	22 (73%)	0.738	1 (3%)	0.049
52	Lidén et al., 2007	HT	32	7 (22%)	23 (72%)	1 (3%)	1 (3%)	25 (78%)	0.785	2 (6%)	0.077
55	Lui et al., 2012	BPTB	14	10 (71%)	4 (29%)	0 (0%)	0 (0%)	4 (29%)	0.310	0 (0%)	0.034
55	Lui et al., 2012	HT	38	32 (84%)	6 (16%)	0 (0%)	0 (0%)	6 (16%)	0.169	0 (0%)	0.013
58	Lund et al., 2014	BPTB	18	14 (76%)	3 (16%)	1 (8%)	0 (0%)	4 (22%)	0.243	1 (6%)	0.081
58	Lund et al., 2014	QT	21	16 (77%)	4 (19%)	1 (4%)	0 (0%)	5 (24%)	0.256	1 (5%)	0.070
70	Noh et al., 2011	HT	33	27 (82%)	5 (15%)	1 (3%)	0 (0%)	6 (18%)	0.194	1 (3%)	0.045
70	Noh et al., 2011	AT	32	26 (81%)	3 (9%)	3 (9%)	0 (0%)	6 (19%)	0.200	3 (9%)	0.108
76	Pinczewski et al., 2007	BPTB	53	43 (81%)	10 (19%)	0 (0%)	0 (0%)	10 (19%)	0.196	0 (0%)	0.009
76	Pinczewski et al., 2007	HT	58	46 (79%)	11 (19%)	1 (2%)	0 (0%)	12 (21%)	0.214	1 (2%)	0.026
82	Sajovic et al., 2018	BPTB	24	19 (79%)	5 (21%)	0 (0%)	0 (0%)	5 (21%)	0.224	0 (0%)	0.020
82	Sajovic et al., 2018	HT	24	19 (79%)	5 (21%)	0 (0%)	0 (0%)	5 (21%)	0.224	0 (0%)	0.020
97	Yadav et al., 2025	HT	36	15 (42%)	12 (33%)	8 (22%)	1 (3%)	21 (58%)	0.589	9 (25%)	0.260
97	Yadav et al., 2025	QT	33	24 (73%)	6 (18%)	3 (9%)	0 (0%)	9 (27%)	0.284	3 (9%)	0.104
98	Yoo et al., 2017	HT	68	33 (49%)	31 (46%)	3 (4%)	1 (1%)	35 (51%)	0.518	4 (6%)	0.066
98	Yoo et al., 2017	TA	64	30 (47%)	28 (44%)	4 (6%)	2 (3%)	34 (53%)	0.535	6 (9%)	0.101

Supplementary Table 9 Graft failure

ID	Study	Subgroup	Sample size	Re-rupture	Clinical failure		Revision**	Event	Proportion
					Pivot-shift†	Lachman‡			
2	Aglietti et al., 2004	BPTB	60	–	0	–	–	0	0.008
2	Aglietti et al., 2004	HT	60	–	0	–	–	0	0.008
5	Anderson et al., 2001	BPTB	35	–	15	–	–	15	0.437
5	Anderson et al., 2001	HT	70	–	27	–	–	27	0.390
6	Anz et al., 2023	BPTB	16	0	–	–	–	0	0.030
6	Anz et al., 2023	HT	16	0	–	–	–	0	0.030
11	Beynnon et al., 2002	BPTB	22	–	0	2	–	2	0.111
11	Beynnon et al., 2002	HT	22	–	6	13	–	19	0.867
12	Bi et al., 2018	HT	62	–	1	–	–	1	0.024
12	Bi et al., 2018	PLT	62	–	0	–	–	0	0.008
13	Björnsson et al., 2016	BPTB	60	–	12	7	–	19	0.322
13	Björnsson et al., 2016	HT	86	–	13	21	–	34	0.399
23	Ejerhed et al., 2003	BPTB	32	–	–	2	–	2	0.077
23	Ejerhed et al., 2003	HT	33	–	–	0	–	0	0.015
25	Eriksson et al., 2001	BPTB	42	–	0	–	–	0	0.012
25	Eriksson et al., 2001	HT	47	–	0	–	–	0	0.011
26	Feller & Webster, 2003	BPTB	42	–	0	0	–	0	0.012
26	Feller & Webster, 2003	HT	47	–	0	0	–	0	0.011
32	Guglielmetti et al., 2021	BPTB	31	–	2	2	–	4	0.143
32	Guglielmetti et al., 2021	HT	31	–	1	3	–	4	0.143
33	Harilainen et al., 2006	BPTB	37	–	0	1	–	1	0.040
33	Harilainen et al., 2006	HT	37	–	1	0	–	1	0.040
34	Heijne & Werner, 2010	BPTB	30	–	3	–	–	3	0.115
34	Heijne & Werner, 2010	HT	29	–	8	–	–	8	0.288
36	Horstmann et al., 2022	HT	26	–	0	–	–	0	0.019
36	Horstmann et al., 2022	QT	23	–	0	–	–	0	0.021
37	Ibrahim et al., 2005	BPTB	40	–	0	0	–	0	0.012
37	Ibrahim et al., 2005	HT	45	–	0	0	–	0	0.011
41	Jansson et al., 2003	BPTB	43	–	0	0	–	0	0.011
41	Jansson et al., 2003	HT	46	–	0	0	–	0	0.011
43	Kautzner et al., 2015	BPTB	74	0	–	–	–	0	0.007
43	Kautzner et al., 2015	HT	73	2	–	–	–	2	0.034
45	Konrads et al., 2016	BPTB	24	–	–	0	–	0	0.020
45	Konrads et al., 2016	HT	23	–	–	2	–	2	0.106
46	Laoruengthana et al., 2009	BPTB	15	–	0	–	–	0	0.032
46	Laoruengthana et al., 2009	HT	13	–	0	–	–	0	0.037
47	Lawhorn et al., 2012	HT	54	–	1	–	–	1	0.028
47	Lawhorn et al., 2012	TA	48	–	0	–	–	0	0.010
49	Laxdal et al., 2005	BPTB	40	1	–	–	–	1	0.037
49	Laxdal et al., 2005	HT	78	2	–	–	–	2	0.032
51	Li et al., 2015	HT	32	–	0	2	–	2	0.077
51	Li et al., 2015	TA	32	–	1	4	–	5	0.169
53	Lind et al., 2020	HT	49	–	0	1	–	1	0.030
53	Lind et al., 2020	QT	47	–	0	2	–	2	0.053
58	Lund et al., 2014	BPTB	18	–	–	1	–	1	0.081
58	Lund et al., 2014	QT	21	–	–	1	–	1	0.070
54	Lucidi et al., 2025	BPTB	19	7	–	–	–	7	0.385
54	Lucidi et al., 2025	HT	20	3	–	–	–	3	0.171
66	Mohtadi et al., 2016	BPTB	110	19				19	0.176

** Revision surgery was defined as subsequent ACLR on the ipsilateral limb after clinical failure or graft re-rupture.

†† Clinical failure was defined by grade 2 (clunk) or 3 (gross) on pivot-shift test

‡‡ Clinical failure was defined by grade 2 (5-10 mm) or 3 (> 10 mm) on Lachman test

66	Mohtadi et al., 2016	HT	110	24			24	0.222
70	Noh et al., 2011	HT	33		1	1	2	0.075
70	Noh et al., 2011	AT	32		3	3	6	0.200
76	Pinczewski et al., 2007	BPTB	58	7	0	0	7	0.128
76	Pinczewski et al., 2007	HT	53	12	0	1	13	0.252
77	Popovic et al., 2024	BPTB	38	3		9	12	0.325
77	Popovic et al., 2024	HT	33	4		5	9	0.284
82	Sajovic et al., 2018	BPTB	24		0	0	0	0.020
82	Sajovic et al., 2018	HT	24		0	0	0	0.020
85	Smith et al., 2020	BPTB	32	0			0	0.015
85	Smith et al., 2020	HT	32	2			2	0.077
87	Sonnery-Cottet et al., 2026	BPTB	273	12			12	0.046
87	Sonnery-Cottet et al., 2026	HT	283	12			12	0.044
90	Taylor et al., 2009	BPTB	32	3			3	0.108
90	Taylor et al., 2009	HT	32	4			4	0.138
96	Wipfler et al., 2011	BPTB	29	3			3	0.119
96	Wipfler et al., 2011	HT	25	3			3	0.137
97	Yadav et al., 2025	HT	36	1	1	9	11	0.315
97	Yadav et al., 2025	QT	33	0	0	3	3	0.104
98	Yoo et al., 2017	HT	68		2	4	6	0.095
98	Yoo et al., 2017	TA	64		3	6	9	0.147

Figures

Figure 1 PRISMA flowchart depicting the study identification and selection process

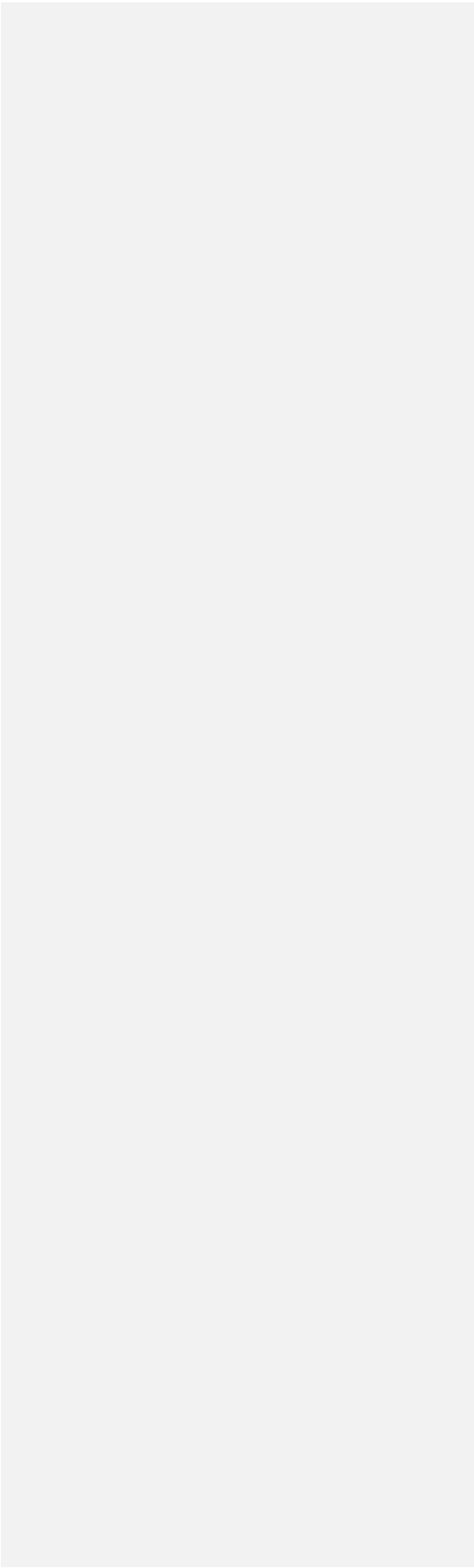


Fig 3. Central Figure

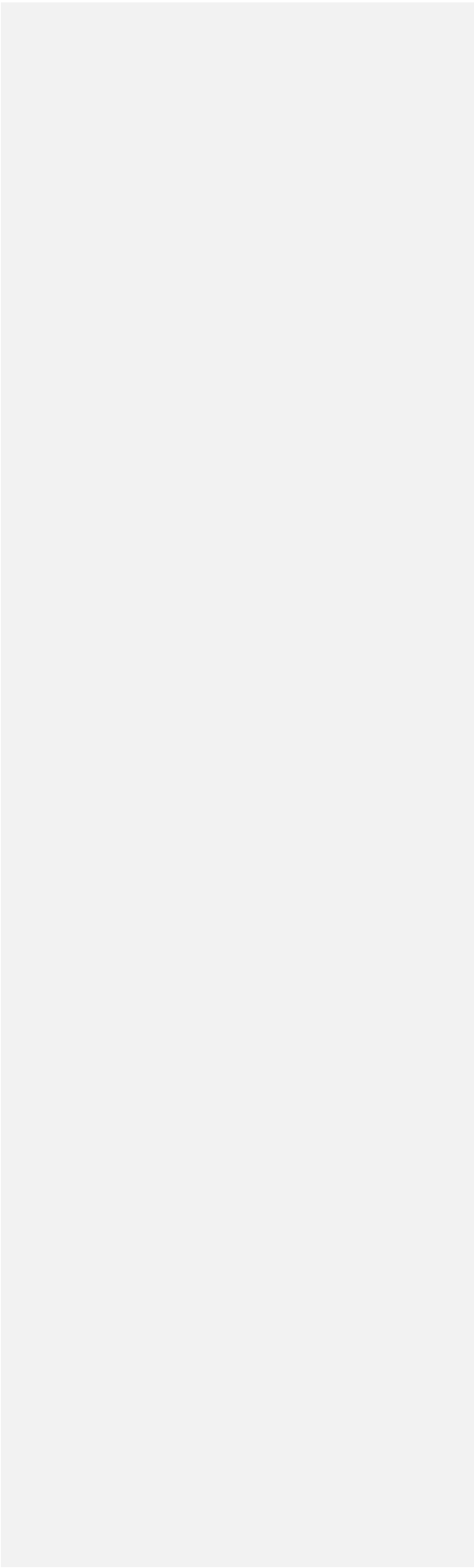


Fig 4. PRISMA flowchart depicting the study identification and selection process

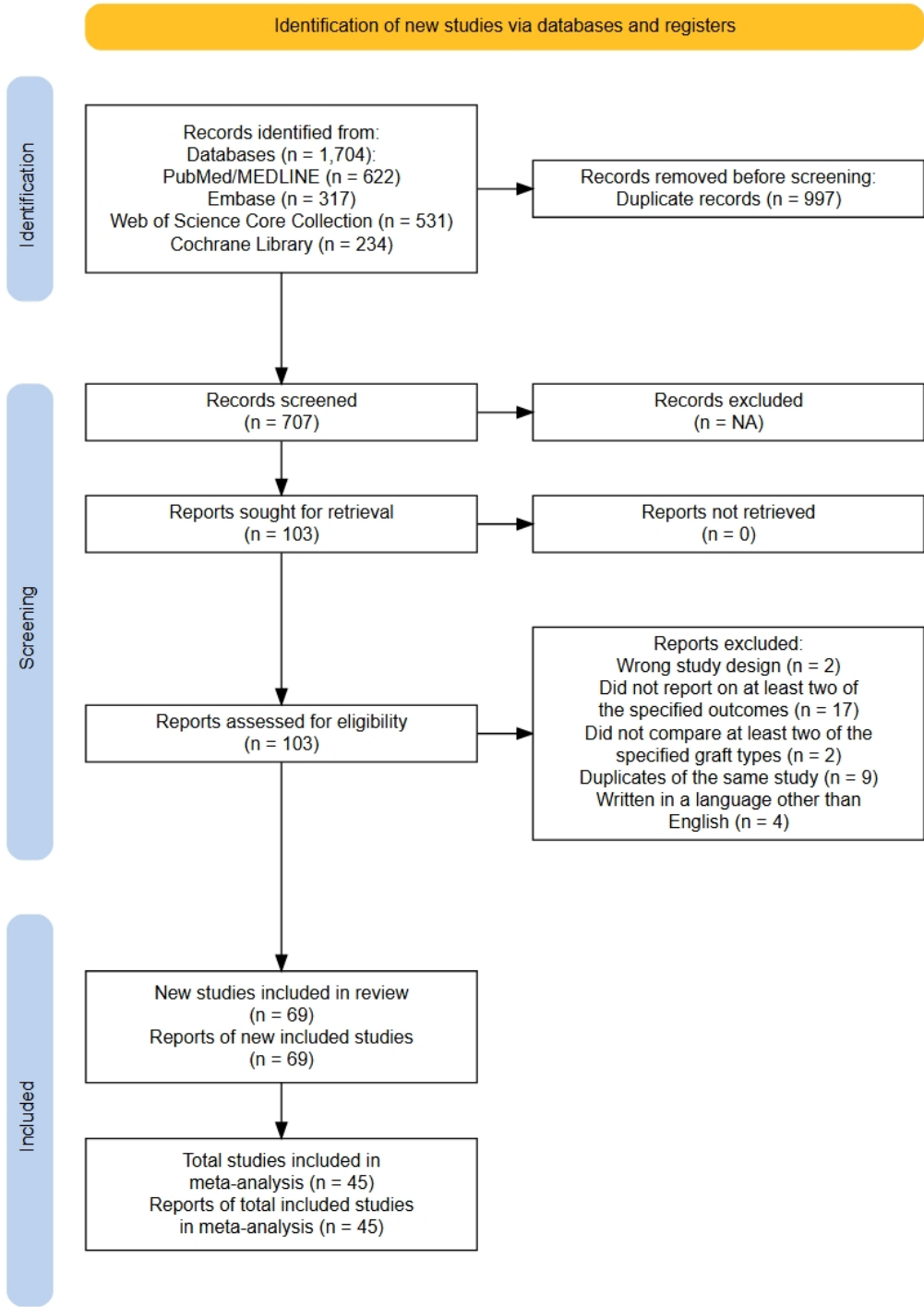


Fig 5. Network graph

